Project 1

Open-Source AI Practice
(Deep Learning Programming Practice)

Kyuhong Shim

Dept. of Intelligent Software

Sungkyunkwan University

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Schedule

Date	Notes	Date	Notes
3. 4.	Orientation	4. 29.	**Project 1 QA**
3. 11.	**Tensor** **Assignment 02**	5. 6.	**Multi-task Learning** **Assignment 07**
3. 18.	Gas accident (no class)	5. 13.	Project 2 (~6/9) Announcement
3. 25.	**Module** **Assignment 03**	5. 20.	**Model Efficiency** **Assignment 08**
4. 1.	**Layer** **Assignment 04**	5. 27.	**Project 2 QA**
4. 8.	**Training** **Assignment 05**	6. 3.	Election day (no class)
4. 15.	Project 1 (~5/5) Announcement	6. 10.	**LLM Prompting** **Assignment 09**
4. 22.	**Optimization/Regularization** **Assignment 06**	6. 17.	Best project presentation

Grading (Updated)

- **Scores**
  - **Assignment** (30%)
  - **Project 1** (35%) – Announcement: 4/15, Due: 5/5
  - **Project 2** (35%) – Announcement: 5/13, Due: 6/9

  - There will be **8 assignments**.
  - Only the **top-6 assignment scores** will count.
  - Each assignment will be graded on a **1-3-5 scale**.
    - If you complete the task correctly, you will get a 5.
    - There is **no report submission** for the assignment.
    - Excellent submissions may be anonymized and shared as an example for others.

Project 1: Semantic Segmentation

- **Semantic Segmentation**
  - Semantic segmetation (task) classifies every pixel into a semantic class.
    - In other words, SS is per-pixel classification.
  - The model can be trained using the standard cross-entropy (CE) loss.
    - The model can be trained using additional losses (e.g., dice loss).

(RGB image)
Provided in JPG

predict

Person
Bicycle
Background

1	0	1	0
1	1	1	1
1	1	1	1
1	2	1	1
1	2	2	1
2	2	2	0
0	0	2	2
0	0	2	2

(Predicted result)
Often saved as PNG

Project 1: Semantic Segmentation

- **mIoU** is the default performance metric.
  - **Mean Intersection-over-Union**
  - Compute IoU per class (including background), and then average the IoU scores.
  - Pixels with the ignore label (e.g., 255 in VOC masks) must be excluded from both the loss and mIoU computation.

$$IoU = \frac{TP}{TP + FP + FN}$$

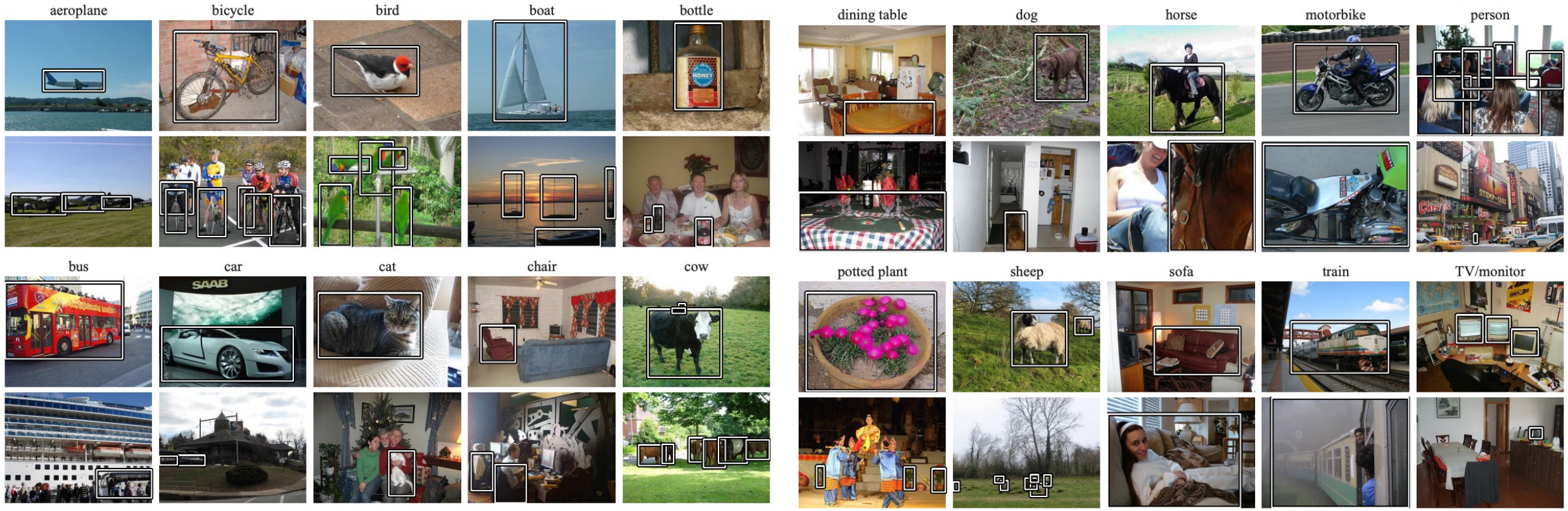
Ground Truth Mask Predicted Mask

Project 1: Data

- **Datasets**
  - You can use **ImageNet-1K, MS-COCO,** and **Pascal-VOC.**
  - These datasets are supported by TorchVision.
    - To use MS-COCO, you may need to install pycocotools.
    - **DO NOT USE THE VALIDATION SET FOR TRAINING.**

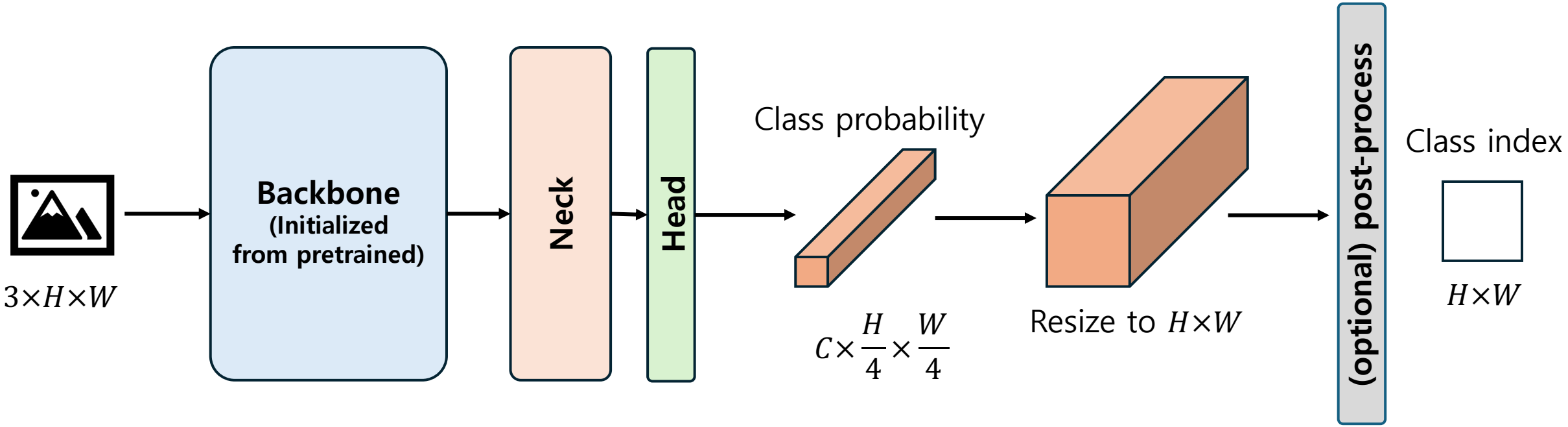
- **Pascal-VOC: 20 classes + 1 background class**
  - You may use VOC training splits from multiple years.
    - However, you must not use any validation or test annotations for training.
  - For evaluation, we will prepare special dataset.
- MS-COCO: 80 classes + 1 background class
  - This dataset is much bigger than Pascal-VOC.
  - All 20 classes in VOC are included in MS-COCO (code), but it needs class name mapping.
    - You may use MS-COCO for additional training.

Project 1: Pascal-VOC



Project 1: Modeling

- **Modeling**
  - You must only use **CNN-based models** (no RNN, no Transformer).
  - A common practice is to use Backbone-Neck-Head architecture.
    - ➢ Full-resolution training is often too expensive, so we reduce the output resolution.
    - ➢ During inference, the result should be resized to the original size.



Project 1: Library

- **Library**
  - You can use:
    - AI: PyTorch and TorchVision
    - Vision: OpenCV, Pillow, Scikit-Image, Matplotlib, etc.
    - Tracking: WandB
  - Do not use:
    - HuggingFace libraries (Datasets, TIMM, Hub, Evaluate, etc.)
    - 3rd party libraries (PyTorch lightning, Accelerate, Albumentations, etc.) that support modeling, data processing, augmentation, training, or evaluation.
    - If you are not sure, ask professor ASAP.

- **AI Usage**
  - You are encouraged to get help from ChatGPT, Claude Code, or Gemini.
    - In your report, briefly describe which AI tools you used and how they helped you.

Project 1: Pretrained Models

- **Pretrained models**
  - **Only use pretrained image classification models that are supported by TorchVision.**
  - TorchVision provides pretrained semantic segmentation models. **Do not use those.**
  - You can use pretrained models that are trained on Imagenet-1K image classification.
    - ➢ ConvNeXt, DenseNet, EfficientNet, MobileNet, ResNet, WideResNet.
    - ➢ Don't use quantized models.

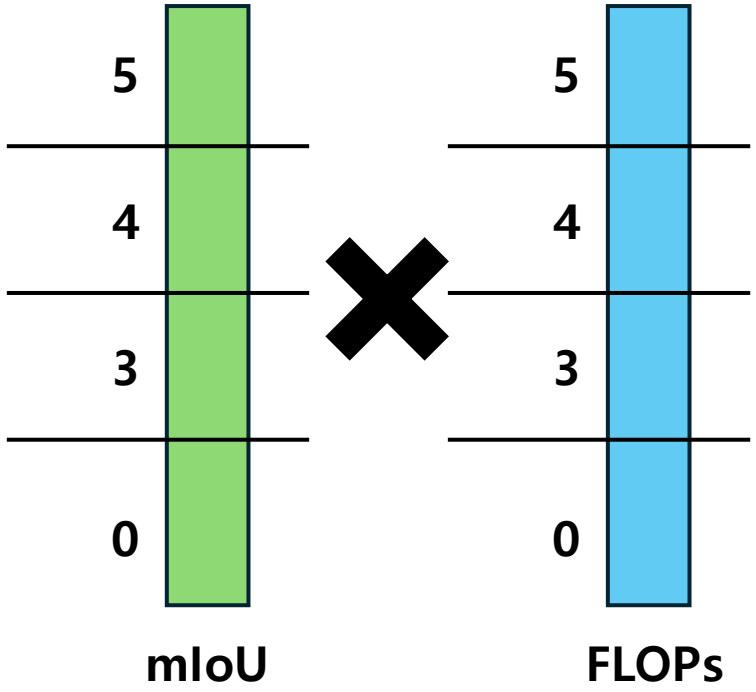
Weight	Mean IoU	pixelwise Acc	Params	GFLOPS	Recipe
DeepLabV3_MobileNet_V3_Large_Weights.COCO_WITH_VOC_LABELS_V1	60.3	91.2	11.0M	10.45	link
DeepLabV3_ResNet101_Weights.COCO_WITH_VOC_LABELS_V1	67.4	92.4	61.0M	258.74	link
DeepLabV3_ResNet50_Weights.COCO_WITH_VOC_LABELS_V1	66.4	92.4	42.0M	178.72	link
FCN_ResNet101_Weights.COCO_WITH_VOC_LABELS_V1	63.7	91.9	54.3M	232.74	link
FCN_ResNet50_Weights.COCO_WITH_VOC_LABELS_V1	60.5	91.4	35.3M	152.72	link
LRASPP_MobileNet_V3_Large_Weights.COCO_WITH_VOC_LABELS_V1	57.9	91.2	3.2M	2.09	link

(Result on MS-COCO subset)
(The table is shown only as a reference, and you must not load these pretrained segmentation weights.)

Project 1: Score

- **The total score will consider multiple factors.**
  - mIoU (performance) @ Closed test set
  - FLOPs (efficiency) @ 3x480x640 input
  - Code quality
  - Report quality

$$S = (S_{mIoU} \times S_{FLOPs}) + S_{code} + S_{report}$$



- **The goal is to balance performance-efficiency.**
  - **Absolute grading, measured on the final model.**
  - mIoU score will be **0-3-4-5** scale.
  - FLOPs score will be **0-3-4-5** scale.

  - Code quality: **5** points.
  - Report quality: **5** points.

Project 1: Score

- **Code quality (not limited to):**
  - Clear module structure
  - Reproducible training and evaluation scripts
  - Proper checkpoint loading and saving
  - Readable implementation
  - No hard-coded test-set assumptions
  - Try to use many things learned in class.
- **Report quality (not limited to):**
  - Model architecture
  - Training recipe
  - Validation results
  - Ablation or trial history
  - Failure case analysis (wrong samples, weak classes, etc.)
  - WandB evidence (monitoring, config, overview, etc.)

Project 1: Submission

Due: **5/5 23:59**

Submission

- Compress the entire codebase, model checkpoint, and report into a single zip file.
- Will announce how to submit your result.

- Important Notes
  - **Test images and submission site will be announced in 4/22 class.**
  - **Threshold will be announced in 4/29 class.**
  - **Only T4 or L4 GPU on Colab are allowed for fair competition.**
    - Report should include WandB log and Overview page capture.
    - We will check your WandB page during class, after project deadline.
  - Report length must be no more than **6 pages**. Use an **11pt font size** for the main text. Submit the report in **PDF** format. There are no other formatting requirements.
  - If you are not sure of anything (code, task, performance, data, etc), ask professor anytime.

Project 1: Submission

(Zip file structure)
2025xxxxxx_project01.zip
- **src/**
- **checkpoints/**
  - model.pth
- **submit/**
  - **img/** **(**submit as empty folder**)**
  - **pred/**
- 2025xxxxx_project01_report.pdf
- pyproject.toml
- README.md

- README.md must include:
  - How to train, install dependencies, and run inference.
  - How can we measure the FLOPs.
  - How to reproduce the submitted result.
    - Take images in "submit/img" folder and predict outpus into "submit/pred" folder.
    - Input/output should have the same file name (0001.**jpg** → 0001.**png**)